

Introduction to Intelligent Robotics (CC3046)
Project Proposal

Autonomous Terrain Navigation Using Sentinel-2 Satellite Imagery with Local Sensing and Dynamic Replanning

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1 Group Identification

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2 Summary of the Goals of the Work

Autonomous navigation in unstructured, real-world terrain remains one of the central challenges in mobile robotics. Classical path-planning algorithms typically assume a fully known global map and a static environment, conditions that rarely hold in practice. This project addresses the more realistic scenario in which a robot must navigate unknown terrain using only local sensor readings, and must continuously update its route as new information is discovered.

The navigable environment is derived from multispectral Sentinel-2 satellite imagery of geographically diverse regions, covering areas that contain forests, rivers, open water, meadows, roads, and urban patches. Using established remote-sensing indices, namely the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Water Index (NDWI), each pixel is classified into a terrain type and assigned a traversal cost that reflects the physical difficulty or environmental impact of crossing that terrain.

A simulated robot agent is deployed on this cost map, equipped with a circular local sensor of configurable radius, analogous to a two-dimensional Light Detection And Ranging (LiDAR) scanner. Crucially, the robot has no prior knowledge of the global map: it perceives only the terrain within its sensor range and must plan a path from a given start position to a goal using this partial information. When the robot discovers terrain that conflicts with its current plan, it triggers a dynamic replanning step, computing a revised path without discarding previously explored cells.

The principal objectives of this work are: (i) to construct realistic terrain cost maps from real Sentinel-2 scenes spanning multiple biomes; (ii) to implement a local-sensing navigation agent with dynamic replanning capability; (iii) to design and conduct controlled experiments comparing global planning baselines against the local-sensing approach; and (iv) to evaluate robustness when artificial noise is introduced into the terrain classification. The total proposal is within the 300-word summary limit.

3 Proposed Approach to Solve the Problem

3.1 Description of the Setup

Simulation Environment

The project is implemented as a Python-based 2D grid simulation. This choice is justified by the nature of the data: Sentinel-2 imagery is a raster dataset in which each pixel maps directly to a grid cell. The simulation is rendered using Matplotlib for static result plots and Pygame for real-time animated visualisation of the robot agent, its sensor footprint, the discovered map, and the planned path. This environment was proposed by the students and approved by the course instructors; the additional complexity of fusing real satellite remote-sensing data with autonomous navigation compensates for not using Webots.

Terrain Cost Map

Eight to ten Sentinel-2 Level-2A scenes are downloaded from geographically diverse locations (examples: a river delta in Portugal, boreal forest in Scandinavia, agricultural plains in Central Europe, semi-arid savanna in sub-Saharan Africa). Each scene is cropped to a 512 x 512 pixel patch at 10 m/pixel native resolution, giving a navigable area of approximately 5.1 km x 5.1 km. Terrain classification and traversal cost assignment rules are as follows:

- NDVI > 0.5: Dense forest. Traversal cost = 8 (high environmental impact, physically difficult terrain).
- $0.2 < \text{NDVI} \leq 0.5$: Sparse vegetation or meadow. Traversal cost = 2 (moderate difficulty).
- NDWI > 0.3: Open water. Traversal cost = ∞ (impassable without bridge; river crossing modelled as a cluster of cells with cost = 50 to allow bridge simulation in extended experiments).
- Remaining pixels (bare soil, roads, urban): Traversal cost = 1 (lowest cost, preferred by the planner).

The resulting cost grid is stored as a NumPy two-dimensional array and used as the substrate for all path-planning algorithms.

Robot Model and Sensor Parameters

The robot is a point agent occupying a single grid cell at each time step, with movement restricted to the eight cardinal and diagonal neighbours (8-connected grid). Its parameters are:

- Position: (row, col) integer coordinates on the cost grid.
- Sensor radius R: configurable integer (default R = 20 pixels = 200 m), defining the circular region within which terrain costs are revealed at each step.
- Known map: a 512 x 512 array initialised to 'unknown' and progressively filled as the robot moves and senses.
- Goal position: a fixed (row, col) coordinate set before each run.

The local-sensing model is the robotic analogue of a 2D LiDAR: instead of measuring distances to obstacles, the sensor reads terrain classification values within radius R around the robot at each step. Unknown cells outside sensor range are treated as free (cost = 1) when computing the initial plan, consistent with an optimistic exploration strategy.

3.2 Algorithms

Algorithm 1 — Dijkstra (Global Baseline)

Dijkstra's algorithm is applied with complete knowledge of the terrain cost map. It guarantees the minimum-cost path from start to goal and serves as the theoretical performance ceiling. No local sensor simulation is involved; the complete cost grid is provided from the outset. This

baseline answers the question: what is the best possible path quality when full information is available?

Algorithm 2 — A* with Euclidean Heuristic (Global Baseline)

A* extends Dijkstra by incorporating an admissible heuristic function. The Euclidean distance to the goal, scaled by the minimum traversal cost (1), is used as the heuristic, which guarantees optimality while substantially reducing the number of cells evaluated. Like Dijkstra, A* operates with global map knowledge. Comparing A* against Dijkstra reveals the computational gain from the heuristic with no loss in path quality.

Algorithm 3 — A* with Local Sensing and Dynamic Replanning (Main Contribution)

This is the central contribution of the project. The robot starts with an empty known map (all cells treated as cost = 1) and runs A* to compute an initial path to the goal. At each step, the sensor reveals terrain within radius R . If any cell on the remaining planned path turns out to have a higher cost than assumed, a replan is triggered: A* is re-run on the current known map, producing a revised path from the current position to the goal. Previously explored cells retain their discovered costs in the known map and are not re-sensed. The process continues until the robot reaches the goal or no feasible path exists.

This approach was selected because it directly exercises the Mapping (TP4) and Motion Planning (TP5) topics of the course, reflects the realistic constraint of partial observability, and produces replanning events that can be systematically measured and analysed.

3.3 Proposed Experiments and Performance Metrics

Experiment 1 — Global vs. Local Planning Comparison

All three algorithms are executed on the same eight terrain scenes with identical start and goal positions. For each (algorithm, scene) pair, 10 independent runs are performed. Performance metrics:

- Path length (metres): total Euclidean distance along the path, computed as the sum of step lengths multiplied by the scene resolution (10 m/pixel).
- Path cost (dimensionless): accumulated traversal cost along the path.
- Path cost ratio (dimensionless): ratio of the local-sensing A* path cost to the Dijkstra optimal cost, measuring sub-optimality due to partial information.
- Computation time (seconds): wall-clock time from algorithm call to path delivery, averaged over 10 runs.

Experiment 2 — Effect of Sensor Radius on Path Quality

The local-sensing A* agent is run on three representative scenes with sensor radii $R \in \{5, 10, 20, 40\}$ pixels, repeated 20 times per configuration. Metrics:

- Number of replannings per run (count): how many times the robot triggered a path revision.
- Path cost ratio vs. global Dijkstra (dimensionless): quality of the discovered path as a function of sensor range.
- Total computation time (seconds): cost of the sensing-planning loop for each radius setting.

Experiment 3 — Robustness to Terrain Classification Noise

Gaussian noise ($\sigma \in \{0.05, 0.10, 0.15\}$ NDVI/NDWI units) is added to the raw index images before cost assignment, simulating atmospheric artefacts and sensor inaccuracies in the satellite data. The local-sensing A* agent is evaluated under each noise level on all eight scenes (10 runs per combination). Metrics:

- Success rate (%): percentage of runs in which the robot reaches the goal within a maximum step budget.

- Path cost deviation (%): mean relative increase in path cost compared to noise-free runs of the same scene.
- Robustness score (dimensionless): fraction of successful runs whose path cost remains within 20% of the noise-free result.

Experiment 4 — Multi-Region Generalisation

The local-sensing A* agent is evaluated on all eight Sentinel-2 scenes representing different biomes. Per-scene performance is reported to identify terrain types where local planning degrades most compared to the global baseline. Metrics per scene:

- Mean path cost ratio vs. Dijkstra (dimensionless).
- Mean number of replannings per run (count).
- Mean path length (metres).

4 Weekly Milestones

The table below outlines the planned tasks week by week from the project start (after Easter break) until the final delivery deadline of 19 June 2025.

Period	Tasks
Week 1 7–11 Apr	• Set up Python project repository; install dependencies (numpy, rasterio, matplotlib, pygame, scikit-image, scipy). • Download and preprocess eight Sentinel-2 Level-2A scenes from diverse geographical locations. • Implement NDVI and NDWI computation pipeline and terrain cost assignment rules. • Visualise raw multispectral bands and generated cost maps for all scenes; validate thresholds visually.
Week 2 14–18 Apr	• Implement 8-connected grid graph data structure with traversal cost lookup. • Implement Dijkstra's algorithm and A* with Euclidean heuristic; validate both on synthetic test grids. • Implement robot agent class: position tracking, step execution, known-map state. • Implement circular local sensor model: reveal terrain within radius R at each step.
Week 3 21–25 Apr	• Implement local-sensing A* with dynamic replanning: detect cost conflicts on current path and trigger replan. • First full integration test: run all three algorithms on one Sentinel-2 scene end-to-end. • Implement Pygame real-time animation: robot position, sensor circle, discovered map overlay, current planned path. • Debug edge cases: impassable water cells, no-path scenarios, sensor boundary conditions.
Week 4 28 Apr–2 May	• Run Experiment 1 (global vs. local comparison) on all eight scenes; store and inspect results. • Implement Gaussian noise injection into NDVI/NDWI images for Experiment 3. • Run Experiment 2 (sensor radius sweep) on three representative scenes. • Begin generating result plots: path overlays on cost maps, cost ratio bar charts.

Week 5 5–9 May	- Run Experiment 3 (robustness to noise) on all eight scenes; compute success rate and cost deviation metrics. - Run Experiment 4 (multi-region generalisation); compile per-scene statistics table. - Finalise all result figures and tables; prepare demo video or live demo for checkpoint presentation. - Prepare 5-minute pitch slides for the 14 May intermediate checkpoint.
Checkpoint 14 May	PROJECT PITCH: demonstrate live or recorded robot navigation on a Sentinel-2 scene, present early Experiment 1 results, and outline remaining work.
Week 6 19–23 May	- Refine experiments and fix issues identified from checkpoint feedback. - Begin writing scientific paper: Abstract, Introduction, and Methodological Approach sections. - Finalise all visualisations; ensure all experiments are fully reproducible from source code.
Week 7 26–30 May	- Write Experimental Evaluation section with all tables, figures, and quantitative discussion. - Write Conclusions and Future Work section. - Internal peer review of paper draft; revise for clarity and IEEE template compliance.
Week 8 2–6 Jun	- Final proofreading and formatting of scientific paper. - Prepare instructions.txt and package source code .zip file. - Prepare and rehearse oral presentation slides (in English). - Submit all deliverables to Moodle by the 19 June deadline.
Final 19 Jun	- FINAL DELIVERY: source code .zip, scientific paper .pdf, and presentation slides submitted to Moodle. - Oral presentation to the jury (10–15 minutes plus Q&A).

5 Objectives for the Intermediate Delivery (Checkpoint — 14 May)

By the intermediate checkpoint on 14 May, the following objectives must be accomplished and demonstrated to the instructors:

1. All eight Sentinel-2 cost maps are generated and visually validated, with correct NDVI and NDWI thresholding and visually sensible terrain class boundaries.
2. The complete robot simulation framework is operational: robot agent, circular local sensor model, and 8-connected grid movement are implemented and functional on real cost maps.
3. All three path-planning algorithms (Dijkstra, global A*, and local-sensing A* with replanning) are implemented and produce correct results on both synthetic test grids and at least one real Sentinel-2 scene.
4. Experiment 1 (global vs. local comparison) is fully executed on all eight scenes; preliminary results are available for at least the path cost and computation time metrics.
5. A real-time Pygame visualisation demonstrating the robot navigating a Sentinel-2 cost map, progressively revealing terrain, and replanning when needed is prepared for live or recorded demonstration.
6. A draft of the scientific paper Introduction and Methodological Approach sections exists.

The 5-minute presentation at the checkpoint will include: a live or recorded demo of the robot navigating a scene, a summary of Experiment 1 preliminary results, and an updated plan for the remaining weeks of the project.